Machine Learning Project
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Analysis and prediction of poisonous mashrooms

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Abstract

Mushroom misidentification presents a substantial health risk, demanding robust methods for accurate classification. This study developed highly accurate machine learning models to determine mushroom edibility using the Kaggle Mushroom Classification dataset.
Preprocessing involved handling missing values in the stalk-root attribute and removing an irrelevant veil-type feature. A J48 decision tree classifier was rigorously evaluated using cross-validation across various feature selection strategies. The odor attribute emerged as the most dominant predictor, exhibiting high Information Gain. All models consistently achieved exceptionally low error rates, with a robust model utilizing multiple highly informative features (including odor, spore-print-color, gill-color, and ring-type) demonstrating near-perfect performance.
These findings underscore the strong predictability of mushroom edibility from morphological features and affirm the effectiveness of compact, informative feature sets for building reliable classification systems, thereby significantly contributing to the prevention of mushroom poisoning.

Keywords: Machine learning, feature selection, error rate, mushroon classification

1. Introduction

Mushroom hunting is not only an enjoyable and stimulating activity, but also one with remarkable benefits. It fully engages the mind, encouraging the development of observational and scientific skills, and its popularity has been steadily increasing. The astonishing diversity of colors, shapes, textures, and sizes of mushrooms can sometimes appear almost unreal. However, this variety also conceals a major risk: the possibility of mistaking a poisonous species for an edible one. According to (1), around 45,000 mushroom species have been identified worldwide, yet only about 2,000 are considered edible. Although mushroom-related deaths are relatively rare (approximately three cases per year, as reported by the National Poison Data System (2)), many individuals experience health problems ranging from mild nausea to severe liver failure due to accidental ingestion of toxic species.

The most effective way to prevent mushroom poisoning is to become familiar with their morphological characteristics. Nevertheless, this is not a trivial task, given the large number of features that need to be considered. In this study, we aim to analyze the most relevant features and build predictive models to determine the edibility of mushrooms based on their appearance using machine learning techniques.

For this purpose, the Mushroom Classification dataset was selected from the Kaggle platform (3). The dataset contains 8,124 instances described by 22 categorical attributes, including morphological and ecological properties such as:

- **Cap_shape**: conical, convex, flat, bell, sunken, knobbed
- **Cap_surface**: grooves, scaly, smooth, fibrous
- **Cap_colour**: buff, cinnamon, grey, green, brown, purple, red, white, yellow, pink
- **Bruises**: yes, no
- **Odour**: anise, creosote, fishy, foul, almond, none, pungent, spicy, musty
- **Gill_attachment**: descending, free, notched, attached
- **Gill_spacing**: crowded, distant, close
- **Gill_size**: narrow, broad
- **Gill_colour**: brown, buff, chocolate, grey, black, orange, pink, purple, red, green, yellow, white
- **Stalk_shape**: tapering, enlarging
- **Stalk_root**: club, cup, equal, bulbous, rhizomorphs, rooted, missing
- **Stalk_surface_above_ring**: scaly, silky, smooth, fibrous
- **Stalk_surface_below_ring**: scaly, silky, smooth, fibrous
- **Stalk_colour_above_ring**: buff, cinnamon, grey, orange, brown, red, white, yellow, pink
- **Stalk_colour_below_ring**: buff, cinnamon, grey, orange, brown, red, white, yellow, pink
- **Veil_type**: universal, partial
- **Veil_colour**: orange, white, yellow, brown
- **Ring_number**: one, two, none
- **Ring_type**: evanescent, flaring, large, cobwebby, none, pendant, sheathing, zone
- **Spore_print_colour**: brown, buff, chocolate, green, black, purple, white, yellow, orange
- **Population**: clustered, numerous, abundant, several, solitary, scattered
- **Habitat**: grasses, leaves, meadows, paths, urban, waste, woods

This paper aims to investigate the relevant mushroom features and evaluate their contribution to the classification task. The effectiveness of these features will be assessed by applying univariate and multivariate filters, as well as manual feature selection, and comparing the performance of the resulting classification models in terms of accuracy.

The remainder of this paper is organized as follows:

1. Data exploration
2. Preprocessing
3. Classification models
4. Evaluation

2. Data Exploration

In the data exploration phase, we analyzed the Mushroom dataset, which contains 8,124 records and 22 categorical features describing mushroom morphology, plus the target class (edible or poisonous). All features are nominal, and only one attribute (stalk-root) contains missing values (approximately 30% of the dataset). The class distribution is relatively balanced, with around 52% edible and 48% poisonous samples.

Univariate analysis showed that some features have very skewed distributions, such as veil-type, which contains only one category and was therefore discarded. Others, such as odor and spore-print-color, displayed clear differences between edible and poisonous mushrooms. In particular, odor alone provides strong predictive information, as edible mushrooms are mostly odorless while poisonous mushrooms often have foul odors.

These findings motivated later steps in the preprocessing pipeline (imputation, feature selection), and helped identify which attributes carry the most discriminative power. Representative visualizations are provided in Figure 1 (distribution of odor by class) and Figure 2 (distribution of spore-print-color).

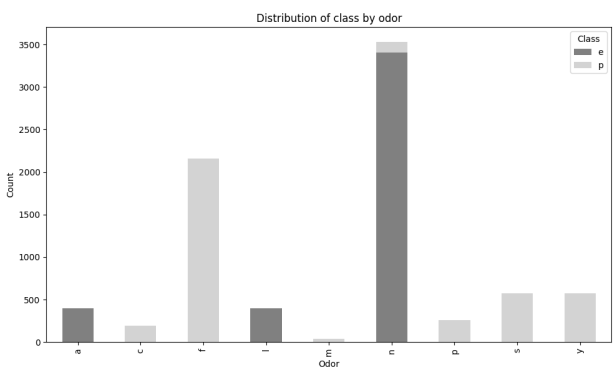

Figure 1. Distribution of `odor` by class (edible vs. poisonous).

3. Preprocessing

The dataset required preprocessing to ensure its suitability for the analysis. In particular, we addressed missing values and removed irrelevant features.

3.1. Missing values

The dataset contains 22 categorical features, but only one of them (stalk-root) has missing values in approximately 30% of the records. To address this issue, we divided the dataset into two subsets: (A) instances with no missing values, and (B) instances with missing values.

Two supervised models, Random Forest and J48, were trained on subset A and evaluated through cross-validation, both achieving perfect metrics. The trained models were then applied to subset B to impute the missing values.

The predictions of the two models were concordant for 2,027 out of 2,460 records, while 453 records received different predictions. For the concordant cases, the agreed value was retained. For the discordant cases, we adopted a probabilistic ensemble strategy: we selected the class with the highest normalized probability, defaulting to Random Forest in case of ties.

The discrepancies between the models arise from the fact that, although both fit the complete data perfectly, in the presence of missing values the attribute combinations are not uniquely determined, leading to different generalization behaviors.

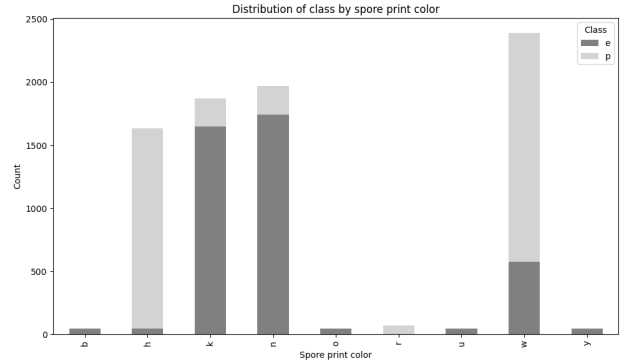

Figure 2. Distribution of `spore-print-color` by class(edible vs. poisonous).

3.2. Removing irrelevant features

During the exploration phase we identified that the feature veil-type is constant across all records. Since it carries no discriminative information, it was removed from the dataset prior to further analysis.

4. Models

For the classification task, we experimented with J48 decision tree classifier. This classifier was chosen because it is well-suited for categorical data such as the Mushroom dataset, require minimal preprocessing, and provide interpretable decision structures.

4.1. Cross validation

To ensure that the obtained performance was not biased by a particular train-test split, all experiments were conducted using k-fold cross-validation (k = 10).

In this procedure, the dataset is randomly partitioned into 10 equal-sized folds. At each iteration, nine folds are used for training and one fold for testing, rotating until each fold has served once as the test set. This ensures that every sample is used both for training and for validation, and the final performance is computed as the average across the folds.

Cross-validation was applied systematically to all models and all feature selection configurations, which guarantees that the results are directly comparable across experimental settings.

4.2. Feature selection

Since the dataset contains 22 descriptive features, several feature selection strategies were applied to investigate whether a reduced subset of attributes could yield comparable predictive performance and to evaluate the impact of dimensionality reduction on robustness. We applied two complementary filtering strategies to identify the most relevant attributes for classification:

- **Univariate filter**: evaluates the predictive power of each attribute independently.
- **Multivariate filter**: considers the combined effect of attribute subsets.

The selected features were:

- **Univariate filter**: odor, spore-print-color, gill-size, habitat, population.
- **Multivariate filter**: odor, gill-spacing, stalk-surface-above-ring, veil-color.

The fact that both filters yielded very similar classification performance (error rate ≈ 0 vs. 0.01) suggests that the dataset can be well classified even with a reduced feature set.

Table 1. Information Gain values for each attribute

Feature Name	Information Gain
odor	0.906
spore-print-color	0.481
gill-color	0.417
ring-type	0.318
stalk-surface-above-ring	0.285
stalk-surface-below-ring	0.272
stalk-color-above-ring	0.254
stalk-color-below-ring	0.241
gill-size	0.232
population	0.202
bruises	0.192
habitat	0.157
stalk-root	0.135
gill-spacing	0.103
cap-shape	0.049
ring-number	0.038
cap-color	0.036
cap-surface	0.029
veil-color	0.024
gill-attachment	0.014
stalk-shape	0.008
veil-type	0.000

Additionally, we examined attribute importance using Information Gain (Table 1). The attribute *odor* exhibited the highest value (0.906), highlighting its dominant role in classification.

While a model trained only on *odor* achieved strong predictive performance (> 0.97 across metrics), relying on a single subjective attribute can introduce measurement bias. To enhance robustness, we also built a model including all attributes with information gain > 0.3, namely:

- odor, spore-print-color, gill-color, ring-type.

This subset provided high performance while distributing predictive power across multiple variables. An overview of the feature selection process is presented in Table 2.

Table 2. Feature selection overview

Method	Selected features
Univariate filter	odor, spore-print-color, gill-size, habitat, population
Multivariate filter	odor, gill-spacing, stalk-surface-above-ring, veil-color
Manual selection	odor
Manual selection (IG 0.3)	odor, spore-print-color, gill-color, ring-type

5. Evaluation

The core of our workflow involved partitioning the dataset into training and testing sets, applying different feature selection strategies, and evaluating the resulting classification models. We investigated three distinct approaches for feature selection: two automated filter methods and two manual selections informed by feature importance.

5.1. Automated Filter Methods

As shown in Table 3, the models trained with features selected by the univariate filter achieved an error rate of 0, while the multivariate filter yielded an error rate of 0.01. These results, consistent across all cross-validation folds, indicate highly effective classification performance even with different feature subsets.

Table 3. Error rates of classification models for different feature selection strategies

Statistics	Univariate	Multivariate	Odor only	Manual selection
Minimum	0.000	0.010	0.015	0.006
Smallest	0.000	0.010	0.015	0.006
Lower Quartile	0.000	0.010	0.015	0.006
Median	0.000	0.010	0.015	0.006
Upper Quartile	0.000	0.010	0.015	0.006
Largest	0.000	0.010	0.015	0.006
Maximum	0.000	0.010	0.015	0.006

5.2. Information Gain Analysis and Manual Feature Selection

As presented in Table 3, the univariate odor-only model exhibited an error rate of 0.015. The accompanying box plot (Figure 3, confidence_odor_only) further illustrates the distribution of a confidence metric for this model, showing values predominantly at 1.0, with a minimum observed value of 0.97 across cross-validation folds. This demonstrates surprisingly strong performance from a single, highly informative feature.

While the odor-only model performed exceptionally well, relying solely on odor presents a practical limitation. Odor is a

subjective measure, prone to potential measurement errors or inconsistencies in real-world data collection, which could compromise the model's robustness.

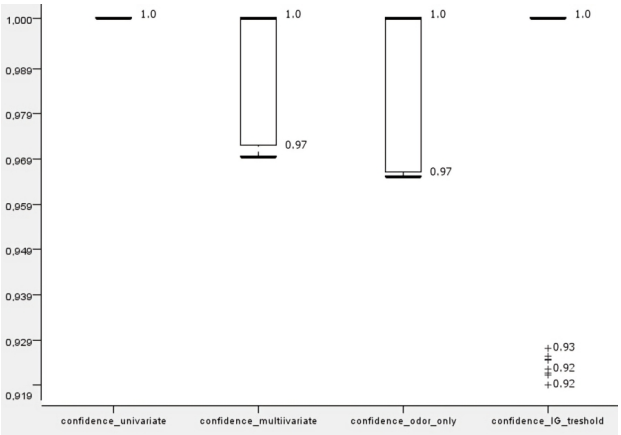

Figure 3. Box plot of confidence metrics for various models.

To mitigate this risk, we constructed a more robust model by selecting all attributes with an Information Gain greater than 0.3.

This final model, using a slightly larger yet still compact set of highly relevant features, achieved an exceptionally low error rate of 0.006 (Table 3, Manual selection). This performance, nearly reaching perfection, indicates a highly accurate and potentially more robust classifier due to its reliance on multiple strong predictors.

The consistent, low error rates across all evaluated feature selection strategies underscore the predictability of the dataset. The minimal performance differences between the models (e.g., 0 error vs. 0.01 or 0.006) suggest that for this specific problem, even a small, carefully selected subset of features can achieve excellent classification accuracy.

6. Conclusion

This study successfully addressed the critical challenge of distinguishing edible from poisonous mushrooms by employing machine learning techniques on the Mushroom Classification dataset. Our primary objective was to analyze the most relevant morphological and ecological features and to build highly predictive models for mushroom edibility, thereby contributing to safer identification practices.

The methodology involved a comprehensive preprocessing pipeline: missing values in the stalk-root attribute were imputed using a probabilistic ensemble of Random Forest and J48 models, demonstrating high concordance in predictions. The non-discriminative veil-type attribute was subsequently removed.

For classification, a J48 decision tree classifier was employed, with all experiments rigorously evaluated through 10-fold cross-validation. Various feature selection strategies were investigated, including univariate and multivariate filters, as well as manual selections guided by Information Gain. Key findings revealed odor as the most significant predictor, exhibiting a remarkably high Information Gain of 0.906.

Evaluation demonstrated exceptional model performance across all feature subsets. Automated univariate and multivariate filters achieved error rates of 0.000 and 0.010, respectively. While an odor-only model also showed strong results (error rate: 0.015), a more robust model was developed by including all features with Information Gain greater than 0.3 (odor, spore-print-color, gill-color, and ring-type), yielding an outstanding error rate of 0.006.

In conclusion, the study successfully built accurate and robust classification models for mushroom edibility, highlighting the critical role of specific features like odor. These findings provide a strong foundation for developing reliable automated systems for safer mushroom identification.

Future work could explore the application of other advanced classification algorithms, such as Gradient Boosting or Support Vector Machines, to compare their performance and robustness against the J48 classifier. Further investigation into the impact of different data imputation strategies, particularly regarding the minor class imbalance observed during the imputation phase, could also refine the overall pipeline.

References

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